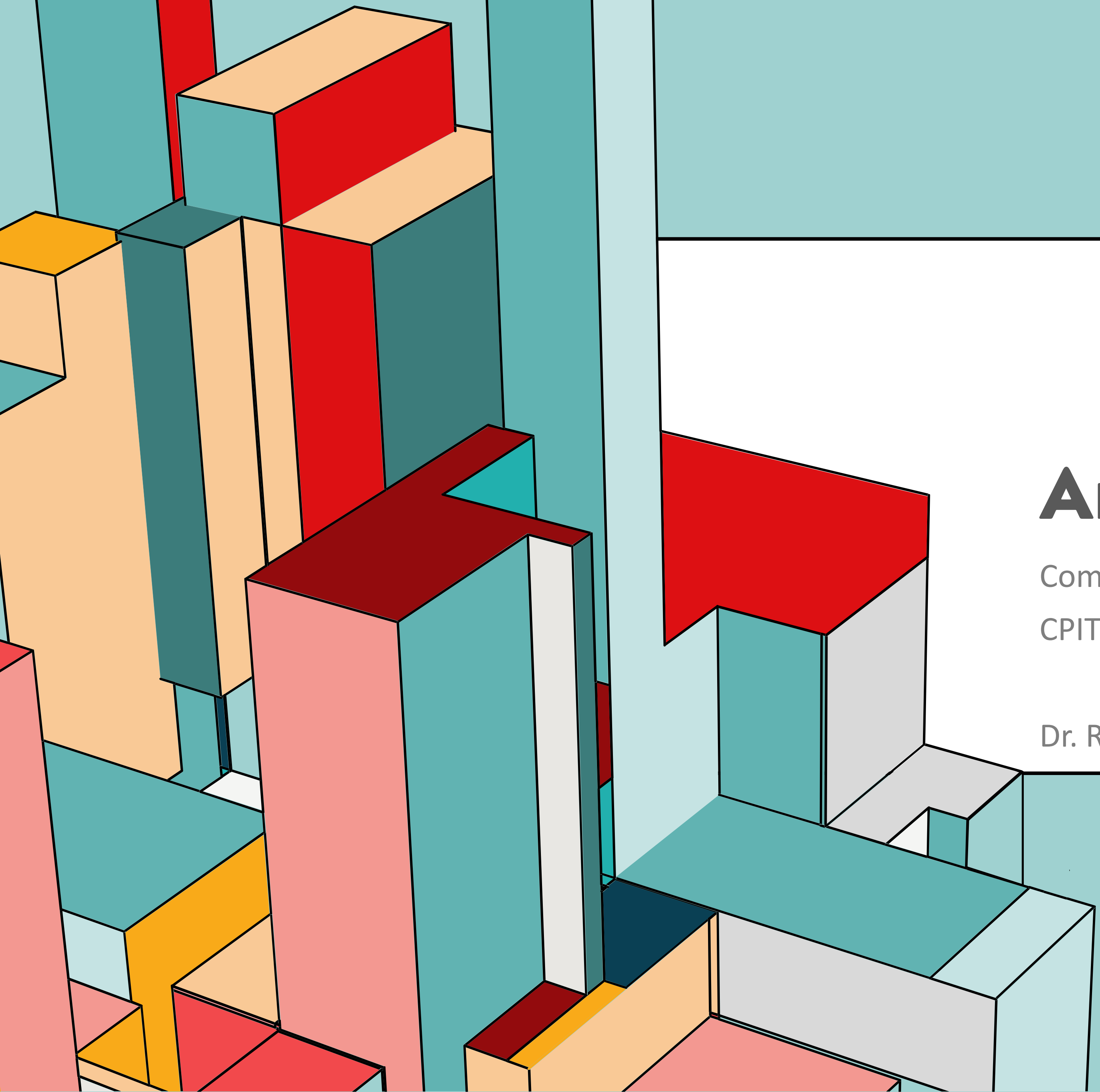


Animation

Computer Graphics

CPIT285

Dr. Rayed AlGhamdi

ANIMATION

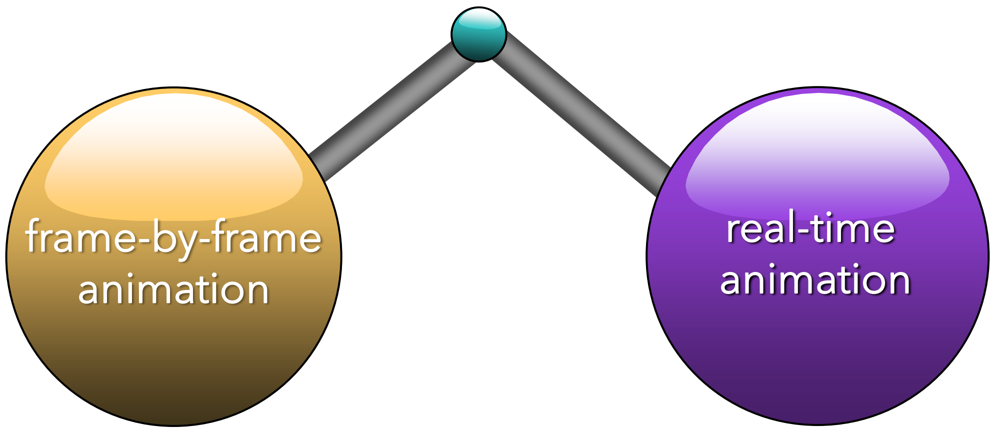
Getting Started

- Objects movement creates interaction in a graphical application
- Animation is an essential component in computer games and videos
- Computer animation generally refers to any time sequence of visual changes in:
    - Translation
    - rotation
    - size
    - color
    - transparency
    - surface texture

ANIMATION

Getting Started

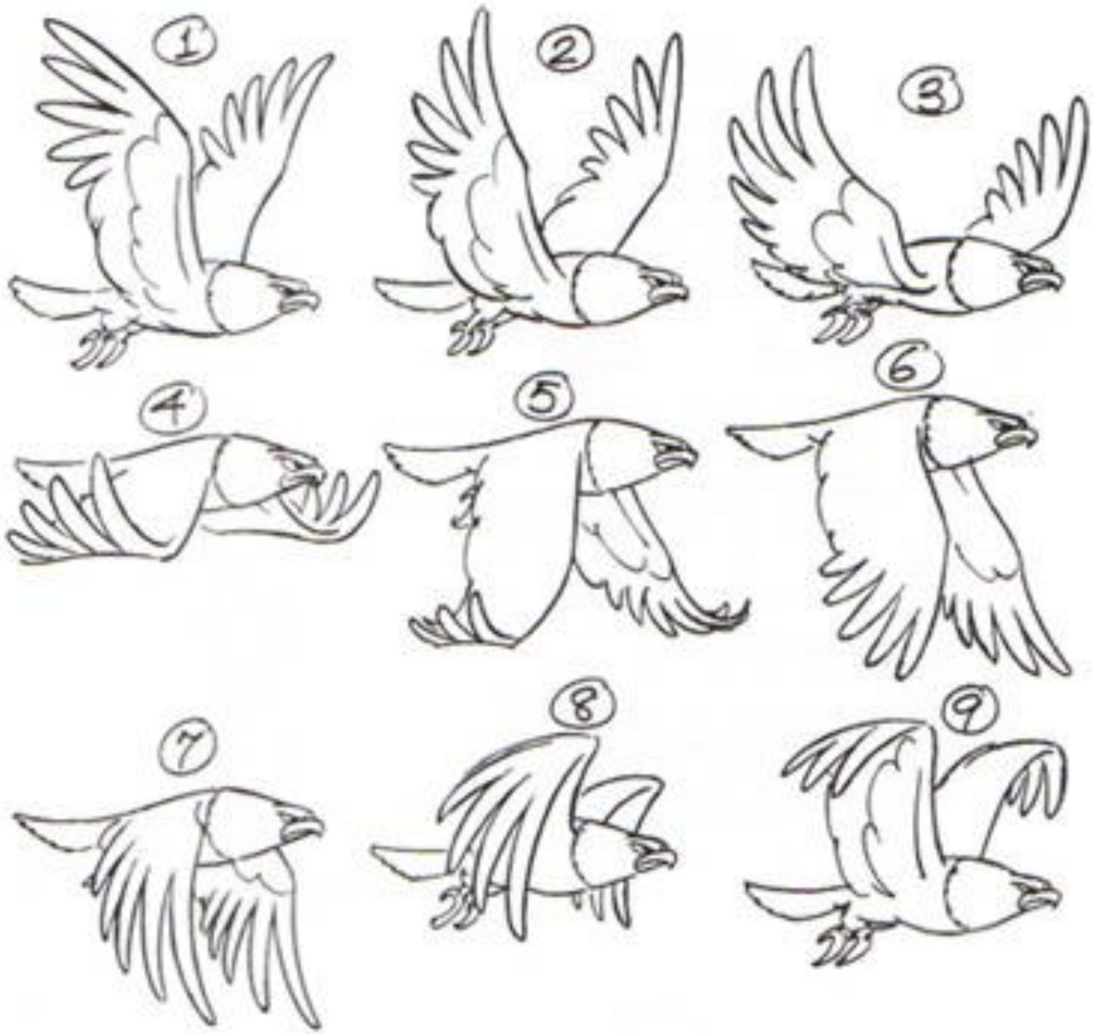
- An animated object can be:
  - dependent on a user inputs
  - Dependent on a timing
  - continuous auto animation
- Generally, objects can be animated based on:



ANIMATION

Animation Methods

- Frame-By-Frame Animation



ANIMATION

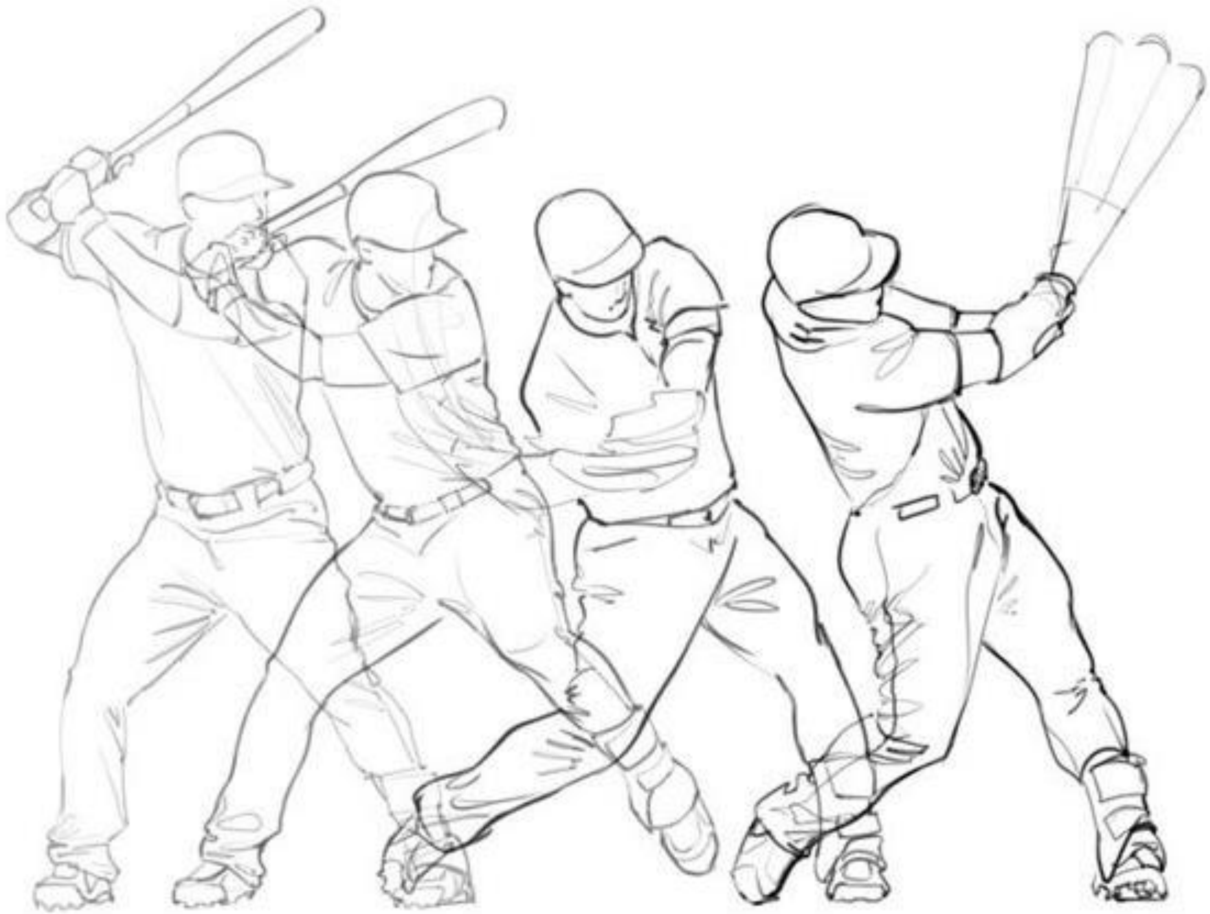
Animation Methods

- Frame-By-Frame Animation
  - frame-by-frame animation is one of the oldest and most popular animation techniques.
  - It is where each incremental frame (image) of animation is drawn individually to create the illusion of movement.
  - This is in contrast to computer-generated or motion graphics animation, where the computer can create images on its own within parameters set by the animator.
  - That’s not to say we don’t use computers for frame-by-frame animation. Back in the day, frame-by-frame was hand-drawn (on paper), so at least now our animators can save time by using computer applications.

ANIMATION

Animation Methods

- Frame-By-Frame Animation



ANIMATION

Animation Methods

- Frame-By-Frame Animation
  - The traditional frame-by-frame animation was handled by drawing each step several times to create the illusion of movement.
  - Keyframing puts the control in the hands of the animator and has roots in hand-drawn traditional animation.
  - With setting up the keyframing, you do not draw/move your object in every single step.
  - All you need to do is to set the starting and ending points and the computer calculates/interpolates or tweens between them.

ANIMATION

Animation Methods

- Frame-By-Frame Animation
  - To experiment with the traditional frame-by-frame animation, use Adobe ImageReady (it is included with the Adobe Photoshop 7 I already shared with you).
  - Download any GIF image and open using Adobe ImageReady to explore each frame and how you can control using layers.

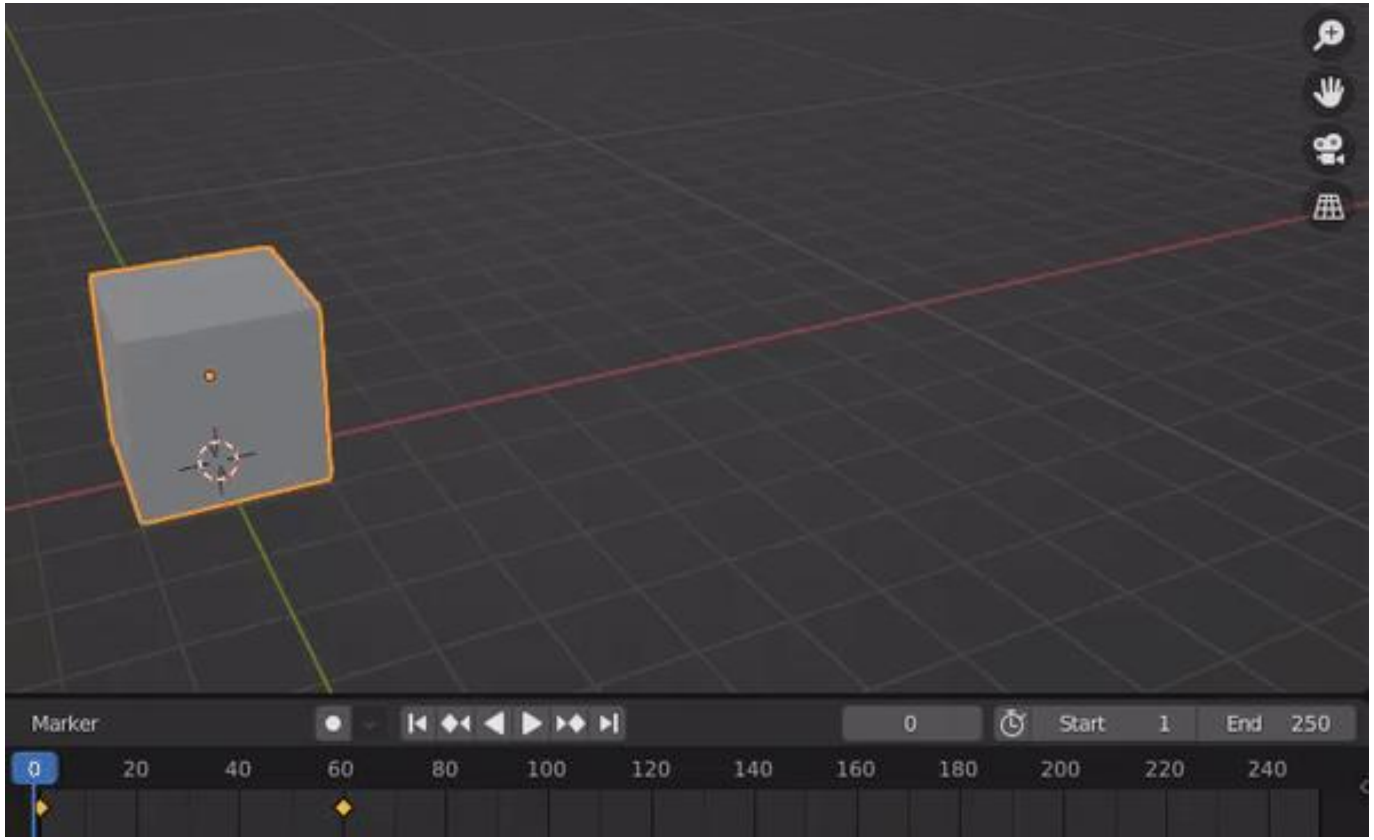
ANIMATION

Animation Methods

- Frame-By-Frame Animation

  Keyframing

  - In this example here, the animator set the first keyframe at zero & the last keyframe at 60 seconds.
  - The computer calculates between these two keyframes to create the frames for you!
  - Try it yourself using Blender, After Effects, or Sony Vegas…

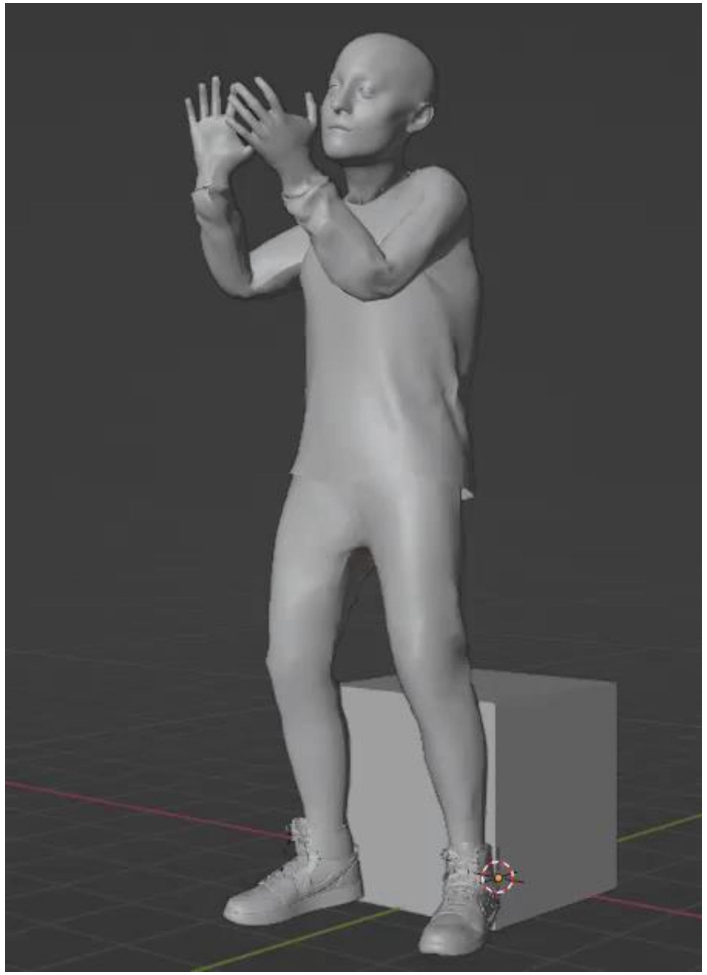


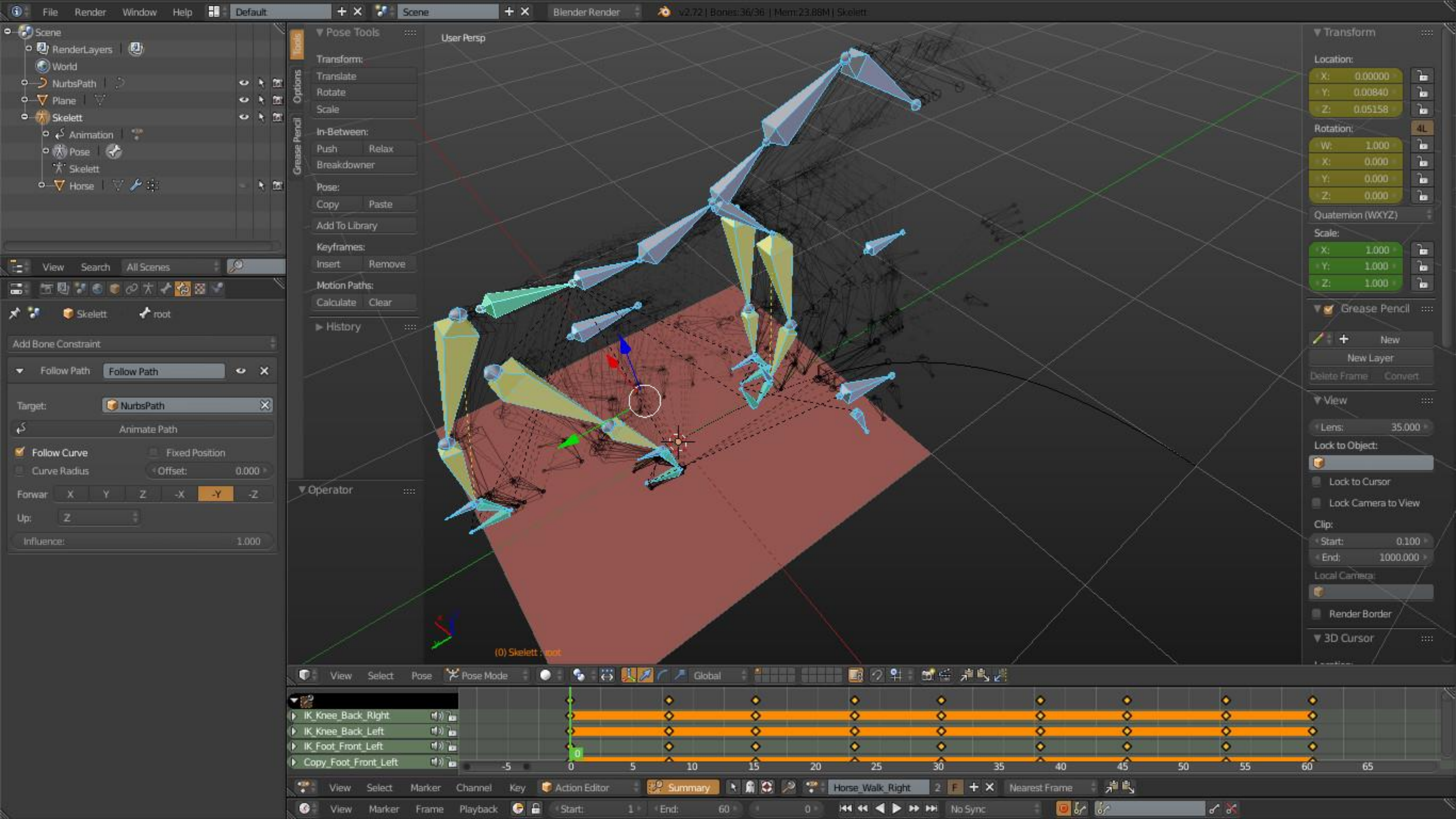
ANIMATION

Animation Methods

Keyframing

- In the previous example (i.e. animating cube), the whole object is animated at once.
- When you want to animate different parts of an object separately, such as legs and arms in a character, you need to rig your object.
- Rigging is the process of creating a skeleton for a 3D model so it can move its parts.
- It simulates the bones of an organism or joints of an object.





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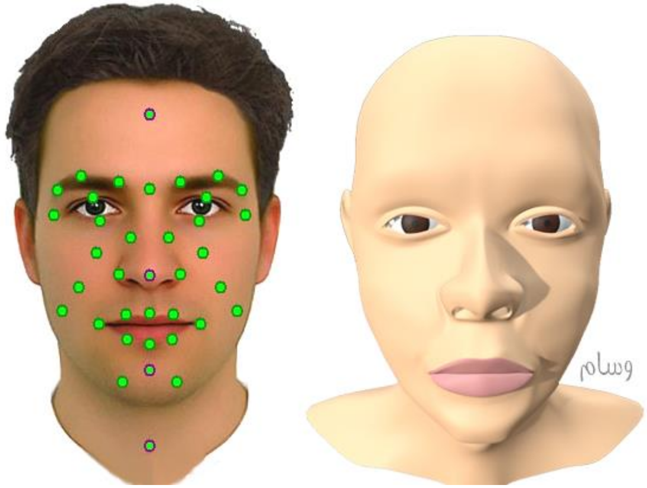
ANIMATION

Animation Methods

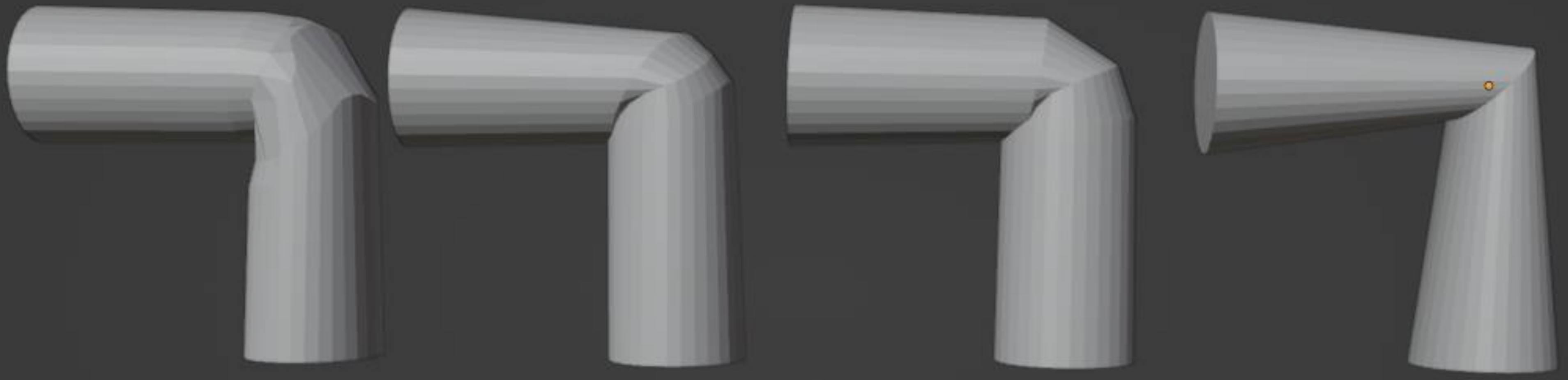
- Real-Time Animation
  - Herby the Elf visits Children at hospital
  - Larry the Zombie - Interactive Animation
  - Facial Motion Capture with Blender



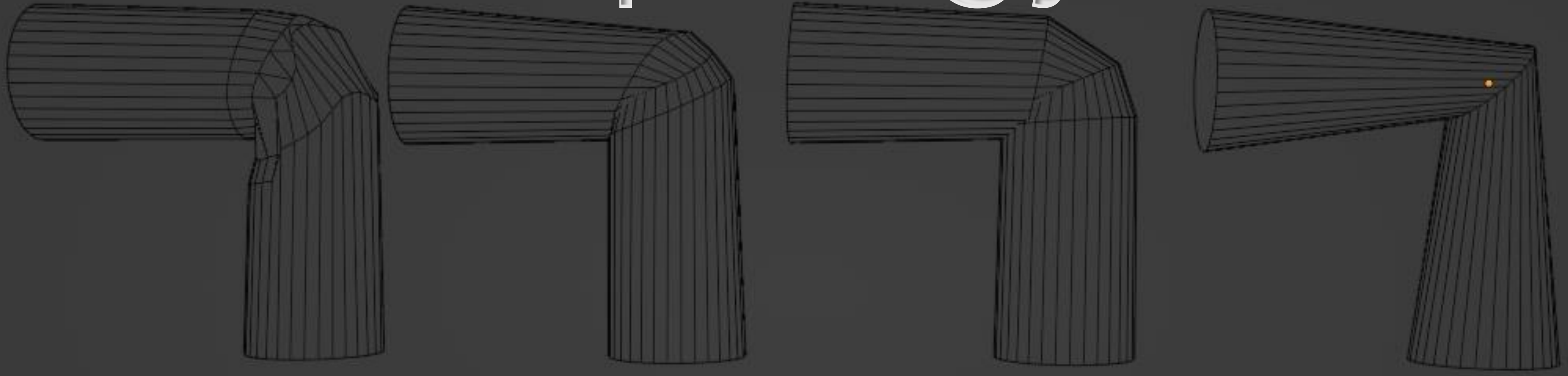
ANIMATION

Animation Methods

- Real-Time Animation
  - Real-Time Animation, also called “Live Animation,” or “Real-Time Performance Capture,” is the process of using a motion capture system to puppet a 3D character live and in real-time.
  - It can be used for previsualization, so a director can preview an animation while the mocap actor is performing, or as a live feed so a real person or audience can communicate with the 3D character in front of them.
  - Real-time mocap animation has the ability to bring 3D characters to life and allows them to be present at game cons, appear on talk shows, speak at keynotes, and cover breaking news.



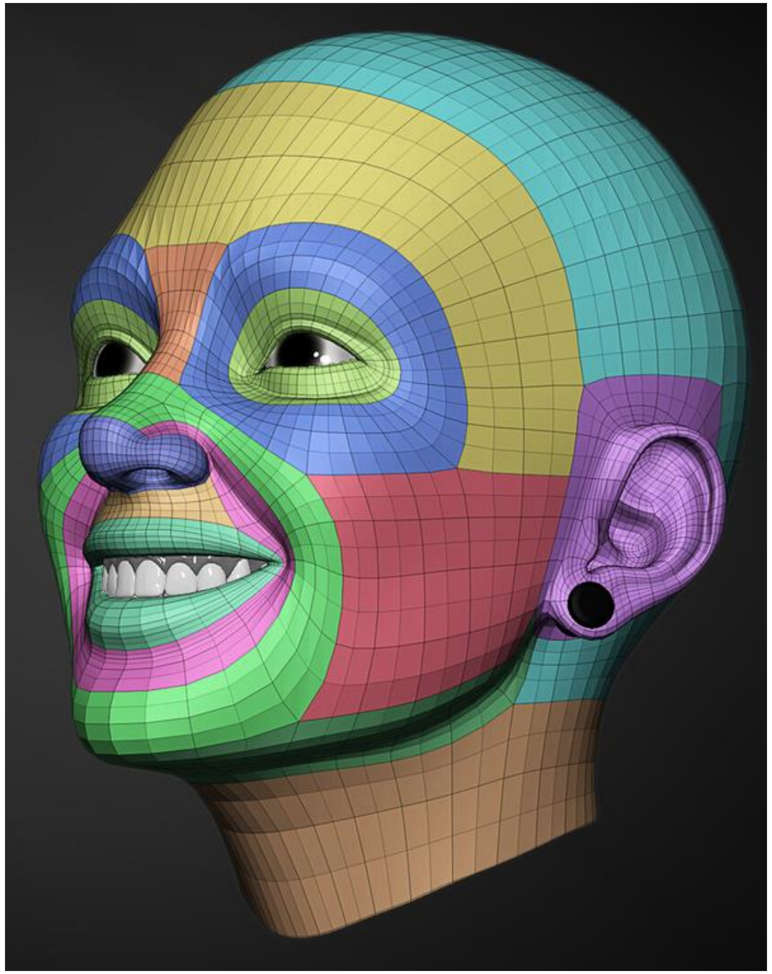
Topology



Topology

What is topology?

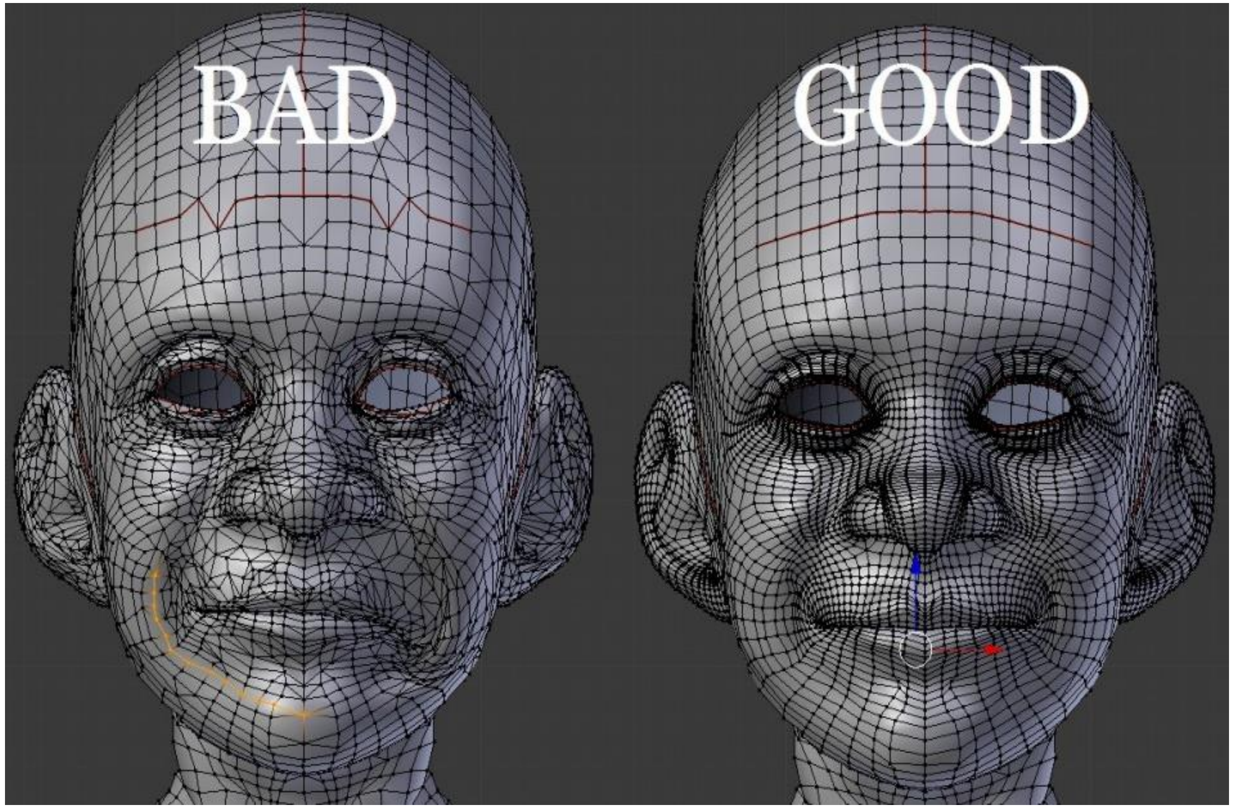
- Modeling 3D objects are made up of hundreds or thousands of simple geometric shapes.
- These geometric shapes are made of a wireframe.
- A wireframe contains many polygons, vertices, edges, arcs, curves, and circles. These shapes form faces in the wireframe design.
- The term topology refers to the organization, flow, and structure of 3D wireframe.

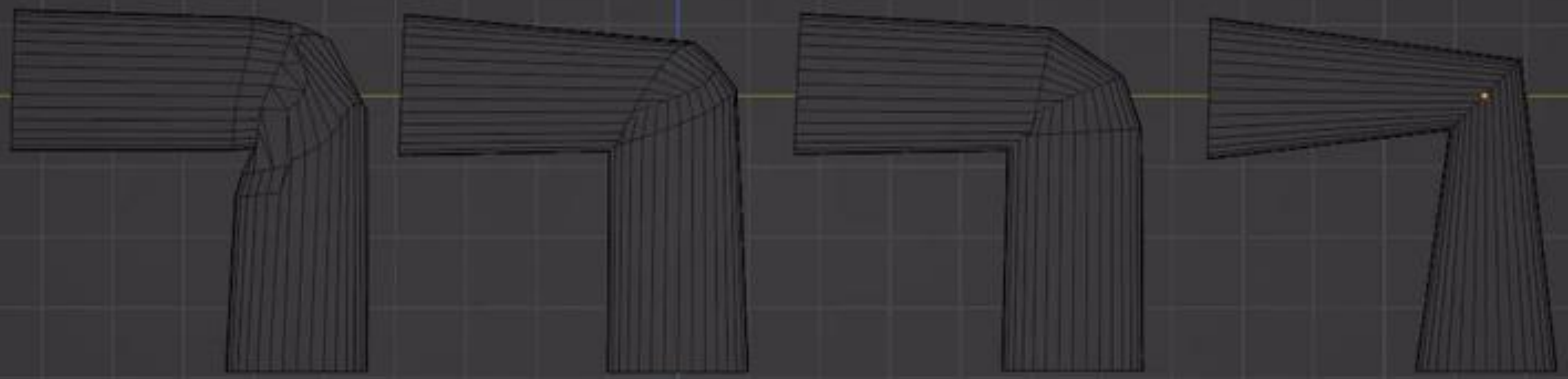


Topology

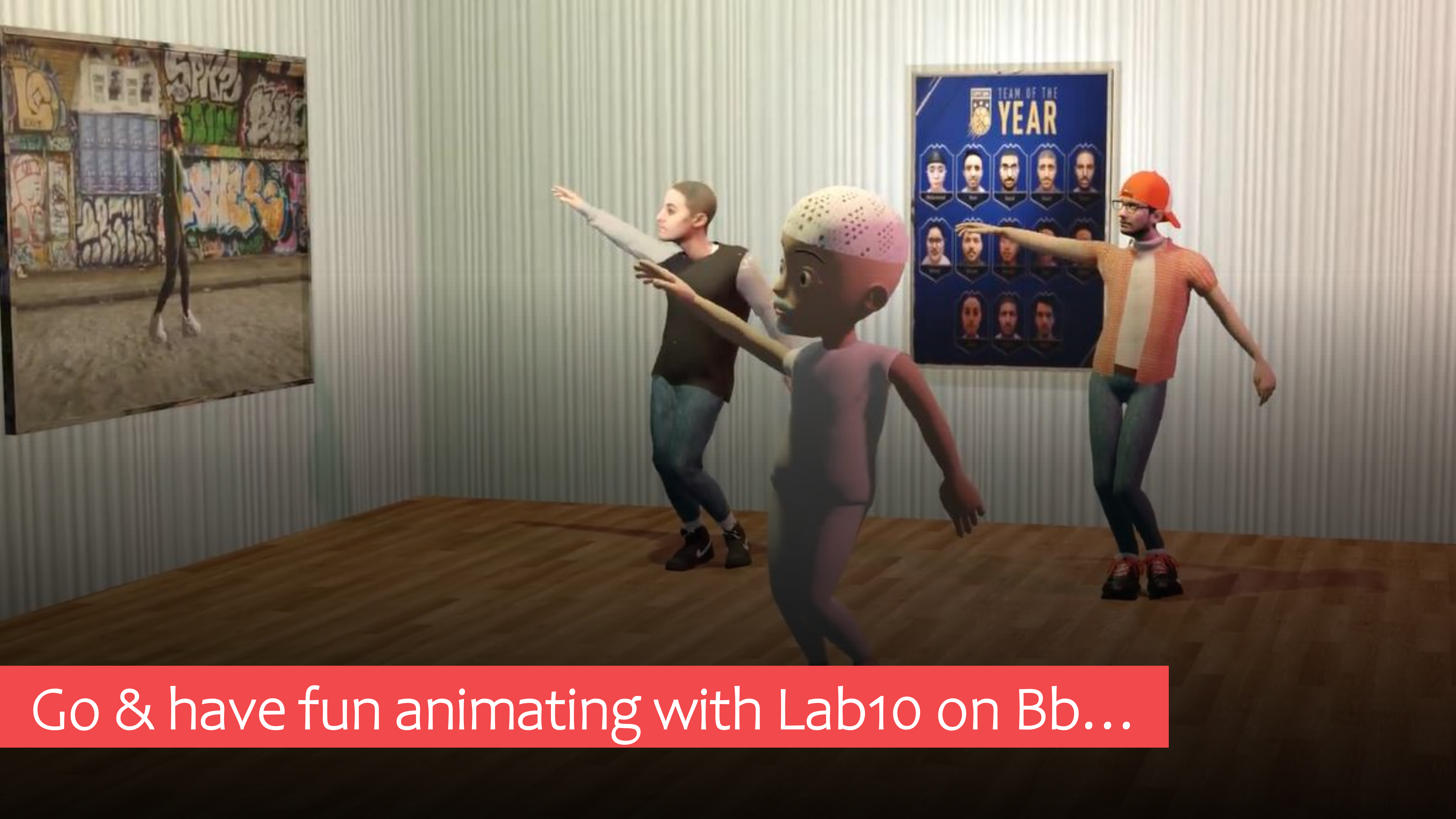
Why do I need to care about topology?

- Clean topology is essential when modeling for 3D animation.
- Maintaining good and clean topology helps to easily make modifications to a model compared to a model with a bad, messy topology!
- It highly helps in making deformed animations look fluid and smooth.





Pay attention to the elbow area and how it differs in each one based on the wireframe organization

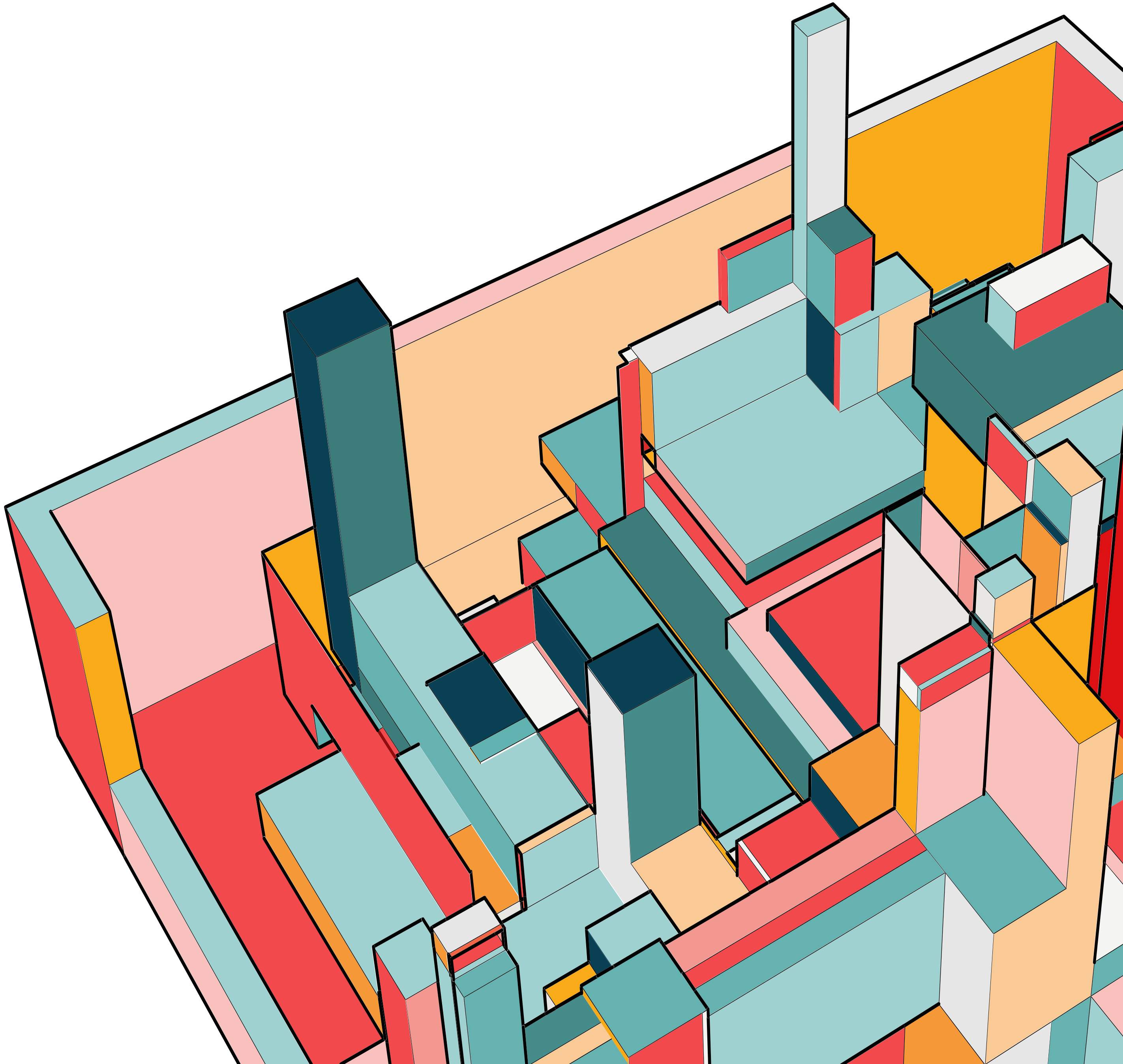


Go & have fun animating with Lab10 on Bb…

100% CONFIDENT

Your journey in this course is enjoyable and beneficial!

Enjoy learning CPIT285!



THANK YOU

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